

Applying DNA barcoding to conservation practice: a case study of endangered birds and large mammals in China

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Abstract Owing to advantages over traditional species identification methods, DNA barcoding is suggested to be a promising tool in conservation research. However, the use of DNA barcoding to accurately identify unknown samples in conservation practices has not been well documented in the literature. To illustrate this issue, we implemented a survey of endangered birds and mammals in China based on mitochondrial Cytochrome c Oxidase subunit I (COI) gene. We included mostly confiscated specimens and non-invasive samples while concealing species information to simulate real-world scenarios of identification. In total, 47 avian and 39 mammalian specimen were re-identified by sequential analyses of online species assignment, genetic distances, phylogenetic reconstruction, and diagnostic nucleotide method. With this multiple analyses approach, 82 individuals were accurately assigned to the species level and four individuals to the genus level. 78.72% of the avian specimen and 87.18% of mammalian specimen identifications were consistent with morphological classification. Among those inconsistent with morphological classification, we

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identified several potential errors including misidentification based on morphology and mislabelling that may have occurred while combining results from different analytical methods. Our case study not only enriches the barcode database, but also reports a successful application of DNA barcoding identification to conservation practices, which could effectively facilitate species identification of unknown samples in conservation practices in the future.

Keywords DNA barcoding · Cytochrome c Oxidase subunit I (COI) · Species identification · Conservation practice · Birds · Mammals

Introduction

Considered a biodiversity hotspot, southwest China hosts a large number of birds and mammals (Myers et al. 2000; Schipper et al. 2008), most of which are endemic and categorized as threatened or endangered species on the IUCN red list (www.iucnredlist.org). Despite legal protection, many species still suffer from poaching and overexploitation (Yiming and Wilcove 2005; Zhang et al. 2008). This pressure stems from the high demand for birds or mammals that are considered to have ornamental value or applications in Chinese medicine, such as the tiger (bone), bear (gallbladder), and forest musk deer (musk) (Feng et al. 2009; Lau et al. 2010; Li and Zhu 2014; Zhou et al. 2004). Rapid and accurate species identification is crucial for conservation management to prohibit the overexploitation and poaching of protected species. While traditional morphological identification practices are often difficult to apply to conservation research due to the scarcity of in-tact specimens, the accurate identification of specimens is essential for administrative officials to justly prosecute violators of endangered species protection laws and thus warrants the need for accurate and efficient molecular methods of species identification.

DNA barcoding has proven to be a powerful tool for rapid species assignment and cryptic species discovery, with promising applications in animal conservation (Hebert et al. 2003, 2004a; Kress et al. 2015; Vernooy et al. 2010). DNA barcoding requires only a trace of DNA for species identification, permitting the use of damaged or incomplete specimens. Furthermore, this molecular procedure is relatively simpler and much faster than that of traditional morphological methods concerning taxonomy, i.e. the identification of new species, thereby accelerating the species identification process (Hebert and Gregory 2005). DNA barcoding also allows those researchers lacking expertise in morphological taxonomy to carry out accurate assessments of unknown specimens (Hebert and Gregory 2005). In last two decades, DNA barcoding of animals has been extensively tested and has gained considerable validation in species authentication of many taxa, including plankton, insects, fishes, birds and mammals (Borisenko et al. 2008; Hajibabaei et al. 2006; Hebert et al. 2004b; Hubert et al. 2008; Nagy et al. 2012; Oba et al. 2015). As a result of recent efforts, researchers have developed a relatively comprehensive reference database of DNA barcodes—laying a solid foundation for the application of DNA barcoding to conservation.

Despite the wide applicability of DNA barcoding in many taxa, the use of DNA barcoding for the identification of unknown specimens in real-world conditions has not been extensively surveyed. Previous DNA barcoding studies have heavily relied on invasive tissue samples, most of which having employed one or two analytical methods for species identification in biodiversity conservation (Ardura et al. 2010; Gonçalves et al. 2015), food

traceability (Quinto et al. 2016; Wong and Hanner 2008), diet analyses (Côté et al. 2013; Jo et al. 2015), bird strikes (Dove et al. 2008; Waugh et al. 2011) and biosecurity etc. (Armstrong and Ball 2005; González-Varo et al. 2014). There are, however, inherent limitations and complementary advantages in these DNA barcoding analyses (DeSalle et al. 2005; van Velzen et al. 2012). It is wise to use multiple analytical methods to get high accuracy in DNA barcoding identification in conservation practices.

Here, we aimed to illustrate how to efficiently employ DNA barcoding for the identification of unknown specimens in practical conservation work, with a case study of many key protected birds and big mammals in China. Multiple DNA barcoding analytic approaches were chosen to achieve high species identification resolution step by step. This case study not only enhances the biotic barcode database, but also provides a guide for the application of DNA barcoding to the identification of unknown samples in conservation practice.

Materials and methods

Sample collection

A total of 94 inventory samples were included (Online Resource Table 1). To mimic authentic conservation scenarios, we focused on confiscated specimens (fur and fragmented or corrupted tissues) and non-invasive samples (feather, fur and faeces). Some blood samples were collected via physical examination. Tissue and faeces samples were stored at -20°C in 95% ethanol ($24 + 3$ individuals). Fifty-six blood samples were stored at -20°C with anticoagulant. Feather (four individuals) and fur samples (seven individuals) were freeze-dried. All samples were empirically identified based on morphological characters by experts from the Nature Museum at Sichuan University, representing 22 species of bird (15 genera, 7 families) and 22 species of mammals (17 genera, 9 families) (MacKinnon et al. 2000; Smith et al. 2010).

These samples represent endangered or rare species that are of great importance to the conservation of biodiversity in China, including 17 species of first-grade, state-protected animals, such as Sichuan Partridge (*Arborophila rufpectus*), black-necked crane (*Grus nigricollis*), south China tiger (*Panthera tigris amoyensis*) and sika deer (*Cervus nippon*), as well as 15 species of second-grade, state-protected animals, such as white-naped crane (*Grus vipio*) and blue sheep (*Pseudois nayaur*). Many of the represented species are also recognized internationally as vulnerable (V), endangered (E) or critically endangered (CE) by the IUCN Red List (version 2015-4).

To simulate real-world scenarios, we concealed species information, using only catalogue numbers to label specimens.

All samples were obtained in accordance with Chinese regulations for the implementation of protection of terrestrial wild animals (State Council Decree [1992] No. 13). All protocols were approved by the Guidelines for Care and Use of Laboratory Animals and the Ethics Committee of Sichuan University (Chengdu, China).

DNA extraction, amplification and sequencing

Genomic DNA was extracted using a standard proteinase K/phenol method with minor modifications (Sambrook and Russell 2001). Fecal DNA extractions were performed using

the QIAamp DNA Stool Mini Kit (Qiagen Inc) under the manufacturer's instructions. DNA quality was verified by 0.8% agarose gel electrophoresis. An approximately 650 bp region near the 5' terminus of the COI gene was amplified as described in Li et al. (2015). For amplification, we employed the universal primers for birds (BirdF1, FalcoFa, BirdR1 and BirdR2) (Kerr et al. 2007) and vertebrates (VF1/VR1 and VF1d/VR1d) (Ivanova et al. 2006; Ward et al. 2005) and designed primers GF_1/GR_1 for Gruidae to maximize amplicon recovery. All primers are listed in Online Resource Table 2.

Amplicons were purified using a DNA gel extraction kit (OMEGA) and bidirectionally sequenced by Invitrogen Trading Guangzhou Co. Ltd. The contigs were assembled with CExpress 9.1, aligned and trimmed to 648 bp after manual examination in MEGA 5.2 (Tamura et al. 2011) and translated into amino acids sequences, to verify that they were free of stop codons and gaps in case of pseudogenes. These COI sequences were submitted to GenBank (Accession Numbers KJ862125–KJ862210).

DNA barcoding identification

Sequences were first run through an online nucleotide BLAST (<http://blast.ncbi.nlm.nih.gov/Blast.cgi>) with the parameter “program selection” set at “highly similar sequences”. The most similar sequence (called “best hit”) with the highest similarity was obtained based upon pairwise alignments by querying individual COI sequences against the nucleotide database. Hits were filtered to those with both coverages and similarities greater than 95%. Queries were then assigned to the species associated with its best hit(s). If there was no such best hit or best hits contained more than one species, the assignment of queries could not be determined. Taking GenBank sequences into account, best hits should exclude the query sequence itself.

Meanwhile, the online BOLD identification system (IDS) was employed to assign the queries (http://www.barcodinglife.com/index.php/IDS_OpenIdEngine) (Ratnasingham and Hebert 2013). We chose “Animal Identification [COI]” database and searched “Species Level Barcode Records”, then input the query sequences in fasta format. This procedure assigns the queries to species or genus level in most cases, showing the most likely candidate species along with top 20 matches while providing tree-based identification. We referred to the trees to infer the taxonomy when the queries were ambiguously identified.

We retrieved conspecific and congeneric COI sequences (Online Resource Table 3) from GenBank according to the preliminary identifications of online analyses, and conducted the following analyses to further discriminate species of the queries.

Kimura two-parameter (K2P) (Kimura 1980) pairwise distances were calculated for birds and mammals in MEGA5.2 (Tamura et al. 2011). Nucleotide gaps were treated as pairwise deletions. Uniform substitution rate was used with substitutions including both transitions and transversions. Genetic distances were calculated at three levels separately: within species, genera and families based on the online identifications, mainly following taxonomic organization suggested by MacKinnon et al. (2000) and Smith et al. (2010). The samples with genetic distances close to the average genetic divergence and interspecific distances generally being greater than their intraspecific distances are considered to be accurately identified, while the specimens with extremely large conspecific genetic distances or very small congeneric distances, for example zero, are thought to be misidentified, although extremely small congeneric distances could reflect real biological phenomena, such as incomplete lineage sorting, hybridization or recent speciation.

Phylogenetic trees were constructed assuming that samples of distinct species would form discrete clusters in a DNA barcode tree. Neighbour joining trees (Saitou and Nei

1987) based on K2P distances of the two datasets were implemented separately in MEGA5.2 (Tamura et al. 2011). We used 1000 bootstrap replicates to assess branch support. Other parameters were set the same as calculating K2P pairwise distances. Typical of DNA barcoding studies, we mainly focused on whether individuals of the same species were clustered together, rather than on the evolutionary relationships between species. We employed the strict tree-based criterion that queries are assigned to the taxon comprising of conspecific members, in which they are nested (Wilson et al. 2011).

To test the robustness of our results and clarify ambiguous identifications, we carried out diagnostic character-based identification with program BLOG2.0 (Weitschek et al. 2013). Two files in fasta format were prepared: one file consisted of a training set of sequences with species assignments and the other contained a testing set of sequences without species information. Classification logic formulas were created from the training set to identify the testing set. The queries were treated so as to maximize the percentage of testing sequences. Parameters for feature selection were set as described in Li et al. (2015). The logic formula with the lowest false positive rate was used for identification. To improve the reliability of the outputs we discarded the GenBank sequences with extremely large intraspecific distances or those failing to cluster with congeneric sequences.

Results

In total, 47 individual birds and 39 individual mammals were successfully sequenced for molecular identification. Three bird sequences and five feline sequences (underlined in Online Resource Table 1) were found to be pseudogenes due to indels and stop codons, then were discarded in subsequent analyses. The amplification rates were 94% for birds and 88.64% for mammals respectively, suggesting strong primer performance.

Online nucleotide BLAST identification

Forty-three avian specimens were identified down to the species level by online BLAST with an effective identification rate of 91.49%. Identification failed for four specimens (see Online Resource Table 4) either because the best hits contained sequences of different species (B03), or because of the low identity of 90% (N9, N10 and N11). However, among the 43 assignments, there were six ambiguous identifications (N1, N2, N17, N23, N24 and N25, Online Resource Table 4a). This ambiguity arose from best hits with identity of 99% representing several closely related species.

Mammalian specimens experienced a 100% effective identification rate, with individuals identified down to species for all specimens. However, the classification of five individuals S29, S30, S36, and S61 (Online Resource Table 4b) was unreliable due to reasons described above regarding the ambiguity of avian specimens.

Online IDS identification

The effective identification rate of online IDS analyses was 95.74% for birds. Among the 45 successful assignments, 31 were assigned to the species level and 14 were classified to genus. The identification of two samples (B10 and B33) failed due to the absence of closely related DNA barcodes in the database (denoted by “no match”). For mammalian

specimens, 97.44% of queries were successfully classified by online IDS, 24 of which were classified to the species level and 14 of which were classified to the genus level. As above, failure to identify one individual (S49) was a result of “no match” in the DNA barcode database.

According to the preliminary identifications, we downloaded COI sequences of related species to conduct the following analyses, including 189 individuals representing 78 avian species and 147 individuals representing 63 mammalian species (Online Resource Table 3). About 64% of avian species and 56% of mammalian species were represented by more than two individuals in the combined dataset.

Genetic distances

K2P pairwise distances of the birds and mammals are summarized in Table 1. Overall, the genetic distances increased as the taxonomic level climbed for both datasets, with true barcode gaps (Meier et al. 2006) being present only in Gruidae, Ciconiidae and Lorisidae. The average K2P pairwise distances of birds within species and genera were 0.40 and 7.64%, respectively. That is, the mean congeneric distance of birds was approximately 19 times the mean conspecific distance. The average intraspecific distance of mammalian specimens was 1.10% with an average interspecific distance of 6.07%, which showed that the average intraspecific genetic divergence of mammals was larger than that of birds, and the gap between the average interspecific and intraspecific genetic distance was generally narrower than that of birds.

Table 1 K2P pairwise distances of COI gene at different taxonomic level for (a) the investigated birds, (b) the investigated mammals

Family	Conspecific distance (%)			Congeneric distance (%)			Confamilial distance (%)		
	Min	Mean	Max	Min	Mean	Max	Min	Mean	Max
(a)									
Gruidae	0	0.17	0.62	1.25	5.59	8.48	5.01	6.58	9.04
Ciconiidae	0	0.23	1.40	2.21	4.84	8.09	8.55	10.17	11.95
Falconidae	0	0.26	2.05	1.57	8.70	11.74	16.36	18.20	19.65
Accipitridae	0	0.69	2.69	0	9.36	15.06	10.07	13.19	17.03
Pandionidae	0	2.23	3.18	–	–	–	–	–	–
Strigidae	0	0.48	3.02	1.09	9.16	17.62	10.27	16.98	22.10
Phasianidae	0	0.22	2.37	0	8.57	14.38	8.52	15.97	22.13
(b)									
Cercopithecidae	0	1.54	6.93	0.15	7.06	14.94	12.56	19.99	26.18
Cebidae	0	1.79	3.02	0	7.03	17.17	20.04	25.02	30.49
Lorisidae	0.31	0.47	0.62	10.53	10.71	10.88	–	–	–
Lemuridae	0	0	0	–	–	–	15.29	15.29	15.29
Cervidae	0	1.55	13.49	1.57	3.53	7.27	4.00	11.96	14.53
Bovidae (Caprinae)	0	0.42	1.89	1.09	2.79	9.81	6.90	12.43	15.73
Felidae (Pantherinae)	0	0.89	6.73	2.20	10.11	12.13	6.59	10.22	13.12
Ursidae	0	0.53	2.21	1.09	6.27	10.93	16.90	20.12	22.70

The italic values stand for genetic distances of taxonomic groups with barcoding gaps

Table 2 The individuals sharing the same DNA barcodes

Species A/individuals		Species B/individuals	
<i>Accipiter virgatus</i>	GQ481252	<i>Accipiter gularis</i>	JF792341
	GQ481253		JF792343
			B03
<i>Syrnaticus ellioti</i>	NC010771	<i>Syrnaticus humiae</i>	NC010774
<i>Saimiri sciureus</i>	NC012775	<i>Saimiri boliviensis</i>	NC021966
			NC018096
<i>Chlorocebus sabaeus</i>	NC008066	<i>Chlorocebus aethiops</i>	AY972687
			AY972651
			DQ069713

Those individuals with overlapping conspecific and congeneric genetic distances (Online Resource Table 5) included three scenarios. First, the individuals had large intraspecific distances (Online Resource Table 5a), most of them only successfully identified as a result of much larger interspecific distances to other samples. The queries with intraspecific distances being generally greater than their interspecific distances were unidentified, including S14, S15, and S18. The downloaded sequence NC016666 labelled as *Cebus apella* in GenBank had an extremely large intraspecific distance of 31.45%, which must be mislabelled, and was removed from the database in the following analyses. Second, the individuals had unexpectedly low interspecific distances (Online Resource Table 5b) and third, the individuals of different species shared the same DNA barcodes with an interspecific distance of zero (Table 2), which failed to be identified here.

Identification by NJ trees

The NJ trees of birds and mammals are showed in Online Resource Figs. 1 and 2, respectively. According to the strict identification criterion of Wilson et al. (2011), 38 out of 47 avian specimens were successfully assigned to species level based on clustered nests containing conspecific samples with bootstrap support values greater than 90%. Taking into account the individuals classified to the genus level (B03, N12, N13 and N14), the efficient identification rate of birds was 89.36%. Specimen B03, belonging to a cluster containing *Accipiter virgatus* and *A. gularis*, shared the same DNA barcode with both species. N12, N13 and N14 formed a sister clade to *Mycteria leucocephala* then clustered with *M. Americana*, and therefore, were assigned to the genus *Mycteria* (Fig. 1). Lacking conspecific DNA barcodes, five specimens (B08, B15, N9, N10 and N11) were only classified to family level. N9, N10 and N11 paralleled to *Mycteria* and nested into a cohesive clade of Ciconiidae (Fig. 1). Similarly, B08 and B15 were identified as Strigidae (Online Resource Fig. 1a).

Thirty-one mammal queries were successfully identified to the species level and five to genus by the DNA barcode tree (Online Resource Fig. 2), with an efficient identification rate of 92.31%. Five individuals (S29, S30, S36, S61 and S64) nested in a monospecific clade containing congeneric samples, respectively. Three samples (S14, S15 and S18) forming a clade with *Panthera onca* and then paralleling to *Uncia uncia* (synonym: *P. uncia*), could only be assigned to Felidae (Online Resource Fig. 2c).

In addition, most of the closely related species with small interspecific distances (*Strix davidi* and *S. uralensis*, *Cervus elaphus* and *C. nippon*) were successfully classified in the

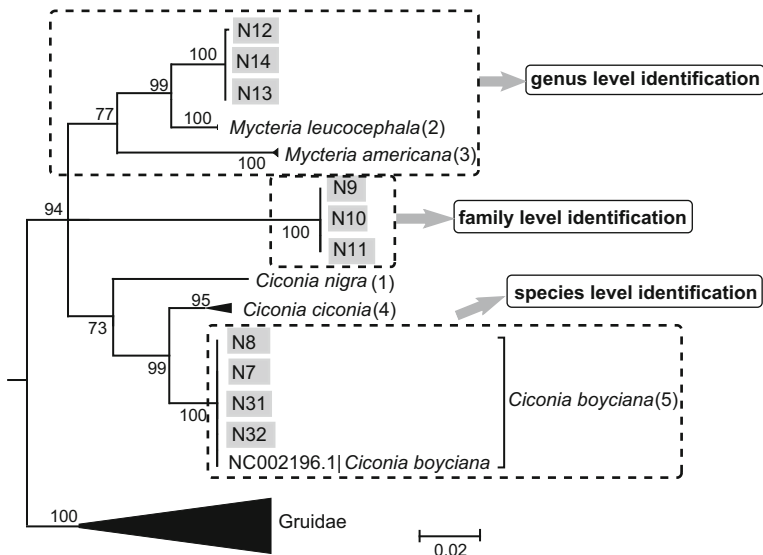


Fig. 1 Neighbour joining tree based on the DNA barcodes of birds (partial shown). The dashed boxes show the three scenarios of **a** species level identification, **b** genus level identification, and **c** family level identification. Our samples are showed in grey shadow. Numbers refer to nodal support values inferred from NJ bootstrap more than 70%

trees by forming distinct and cohesive clades. Meanwhile, some species with large intraspecific distances were also successfully identified by forming monospecific clades (*A. gentilis*, and *Muntiacus reevesi*). However, identification of several species with large intraspecific distances (*C. elaphus*, *Elaphurus davidianus*, *Papio anubis*, and *Chlorocebus aethiops*) failed due to their resolving into paraphyletic groups. Again, NJ trees could not discriminate samples sharing the same DNA barcode.

Identification of BLOG

We conducted BLOG analysis with the combined data to validate our identification results. Most investigated species obtained classification logic formulas (Online Resource Table 6) supporting our previous species assignments. The failed identifications with no diagnostic formulas arose from species sharing DNA barcode, as shown in Table 2.

Combining different DNA barcoding identification results

DNA barcoding identifications based on different analytic methods are summarized in Table 3 and Online Resource Table 7, and compared with the morphologically putative classification. In summary, 82 out of 86 queries were effectively classified to species and four individuals to genus, and 19 queries were successfully classified after interpretation of results from multiple analyses. The DNA barcoding identifications were congruent with morphological classification in 78.72% birds and 87.18% mammals respectively.

For the 15 cases which were incongruent (Table 3), we inferred that seven samples were probably misidentified based on morphological analysis as their sibling species. Two samples were most likely mislabelled as distantly-related species, and four individuals

Table 3 DNA barcoding identification of (a) birds, (b) mammals (partial shown), combined with morphologically putative species assignments

Sample vouchers	Online BLAST	Online BOLD	Distance	NJ tree	DNA barcoding identification	Morphological identification
(a)						
N1, N2, N25	<i>Grus monacha</i>	<i>Grus (Grus monacha)</i>	–	–	–	–
N17, N23, N24	<i>Grus nigricollis</i>	<i>Grus (Grus nigricollis)</i>	–	–	–	–
N7 ^b , N8 ^b , N31 ^b , N32 ^b	<i>Ciconia boyciana</i>	–	–	–	–	<i>Ciconia ciconia</i>
N9 ^a , N10 ^a , N11 ^a	<i>Ciconia maguari</i>	<i>Leptopilos crumenifer</i>	<i>Leptopilos crumenifer</i>	Ciconiidae	<i>Leptopilos crumenifer</i>	<i>Leptopilos crumenifer</i>
N12 ^a , N13 ^a , N14 ^a	<i>Mycteria leucocephala</i>	<i>Mycteria (Mycteria leucocephala)</i>	–	<i>Mycteria</i> sp.	<i>Mycteria</i> sp.	<i>Mycteria ibis</i>
B03 ^a	<i>Accipiter gularis/Accipiter virgatus</i>	<i>Accipiter (Accipiter gularis)</i>	<i>Accipiter</i> sp.	<i>Accipiter</i> sp.	<i>Accipiter</i> sp.	<i>Accipiter virgatus</i>
B05 ^{a,b}	<i>Falco rusticolus</i>	<i>Falco (Falco biarmicus)</i>	–	–	–	<i>Falco cherrug</i>
B08 ^a	<i>Asio otus</i>	<i>Asio (Asio otus)</i>	–	Strigidae	–	–
B13	<i>Falco tinnunculus</i>	<i>Falco (Falco tinnunculus)</i>	–	–	–	–
B15 ^a	<i>Athene noctua</i>	–	–	Strigidae	–	–
B16 ^c	<i>Buteo hemilasius</i>	<i>Buteo (Buteo hemilasius)</i>	<i>Buteo hemilasius</i>	<i>Buteo hemilasius</i>	<i>Buteo hemilasius</i>	<i>Aquila clanga</i>
B10 ^a	<i>Arborophila ardens</i>	No match	–	–	–	–
B33 ^a	<i>Arborophila brunneopectus</i>	No match	–	–	–	–
(b)						
A11	<i>Pseudois nayaur</i>	<i>Pseudois (Pseudois nayaur)</i>	–	–	–	–
A17	<i>Pseudois schaeferi</i>	<i>Pseudois (Pseudois schaeferi)</i>	–	–	–	–
A23	<i>Cervus unicorn</i>	<i>Cervus (Cervus unicorn)</i>	<i>Cervus unicorn</i>	<i>Cervus unicorn</i>	<i>Cervus unicorn</i>	<i>Cervus unicorn</i>
A29	<i>Cervus albirostris</i>	<i>Cervus (Cervus albirostris)</i>	–	–	–	–
M1, Q2, H26	<i>Cervus nippon</i>	<i>Cervus (Cervus nippon)</i>	–	–	–	–

Table 3 continued

Sample vouchers	Online BLAST	Online BOLD	Distance	NJ tree	DNA barcoding identification	Morphological identification
S30, S36	<i>Cercopithecus mitis</i>	<i>Cercopithecus</i> (<i>Cercopithecus mitis</i>)	–	<i>Cercopithecus</i> sp.	–	–
S64 ^a	<i>Chlorocebus aethiops</i>	<i>Cercopithecus aethiops</i>	–	<i>Chlorocebus</i> sp.	–	–
S63 ^b	<i>Trachypithecus hatinhensis</i>	–	–	–	–	<i>Trachypithecus francoisi</i>
S29 ^a	<i>Papio anubis</i>	<i>Papio (Papio hamadryas)</i>	–	<i>Papio</i> sp.	–	<i>Papio</i> sp.
S61 ^a	<i>Papio hamadryas</i>	<i>Papio (Papio hamadryas)</i>	–	<i>Papio</i> sp.	–	<i>Papio</i> sp.
S51 ^c	<i>Saimiri sciureus</i>	–	–	–	–	<i>Cuon alpinus</i>
S49 ^{a,b}	<i>Nycticebus pygmaeus</i>	No match	–	–	–	<i>Nycticebus coucang</i>
S14 ^a , S15 ^a , S18 ^a	<i>Panthera leo</i>	<i>Panthera (Panthera leo)</i>	<i>Panthera</i> sp.	Felidae	–	–

The line-through represents the result same to the online BLAST of each sample. Bold values indicated samples with inconsistent morphological and molecular identification. Underlined samples indicated those just classified to generic level based on DNA barcoding

^a Showed the samples that obtained species assignments by concluding multiple analytical results

^b Showed potential morphologically-misidentified samples

^c Showed mislabelled samples

were only classified to genus based on shared DNA barcode (B03) or lacking conspecific barcodes (N12–N14). Individual S29 and S61, identified to species based on DNA barcoding while morphologically classified as '*Papio* sp.', were also counted in the incongruent cases.

Discussion

A successful application of DNA barcoding identification to conservation

Although DNA barcoding has been applied to a vast array of fauna, most of these studies mainly focused on whether DNA barcoding was suitable for species assignment of certain animal groups (Taylor and Harris 2012). The applications of DNA barcoding to identify unknown specimens in animal conservation practices have yet to be intensively studied. Our research represents a case study of DNA barcoding application to conservation practices based on key protected bird and large mammal species in China, many of which are currently listed as threatened or endangered on the IUCN red list. To better illustrate the efficiency and reliability of barcoding identification in real-world scenarios of conservation, we mimicked the conditions by using real confiscated materials (fur and fragmented or corrupted tissues) or non-invasive samples (feather, faeces), and by concealing species information of samples prior to molecular identification. With multiple DNA barcoding analyses based on COI sequences, 82 out of 86 unknown samples were accurately identified to the species level and four samples to the genus level, most of which were concordant with morphological classifications. Moreover, some individuals, such as S29 and S61 that were only classified to genus level ('*Papio* sp.') through the morphological approach, were successfully identified to species through DNA barcodes. Our case study demonstrated the amenability and reliability of this molecular species identification tool in conservation practices.

Multiple analyses generating reliable DNA barcoding identification

Incomplete or decayed samples of unknown specimens are often difficult or impossible to identify through traditional morphological classification procedures, making species identification one of the biggest challenges in conservation practices. The simple online DNA barcoding identification analysis based on the GenBank or BOLD databases has proven to be efficient and reliable in many studies (Barber and Boyce 2006; Côté et al. 2013; Dalton and Kotze 2011) as well as in our research. However, there was generally no best-performing method i.e., a given method could perform better than another for a given taxonomic scenario, whereas the reverse could be true for another one, see also Austerlitz et al. (2009). To meet the high accuracy requirement in species assignment of unknown specimens in conservation practice, it is better to use a combined analytical approach to conduct DNA barcoding identification. In this case study, we first got a species framework through online BLAST and IDS analysis. Genetic distance and phylogenetic relationship were then assessed from a combination of sample sequences and public barcodes to get detailed information. Lastly, character-based analyses—BLOG was employed to verify the obtained results, especially for weak positive results.

This approach proved effective. Totally, 95.35% unknown samples were accurately identified to species, and approximately 22% of which that failed to be identified by one

method, were successfully classified through the combined analytical approach. Genetic distance and phylogenetic reconstruction were necessary complementary analyses to online identification analyses, given that they helped accurately identify the queries that got different species classification in the two online analyses (Table 3). For example, B10 and B33, unidentified in IDS, were successfully assigned by BLAST, genetic distance, NJ tree and BLOG. N12–N14 were ambiguously classified as *M. leucocephala* with weak support from online analyses, genetic distances and phylogenetic tree, while the BLOG analysis revealed that the diagnostic characters drawn from them (Pos66 = C, Pos399 = A, Pos444 = T, Pos627 = T) were incongruent with that from the downloaded sequences of *M. leucocephala* (HM804932 and HM804933, Pos30 = T, Pos207 = C, Pos399 = G, Pos627 = T), and there were 16 different nucleotide positions between them after realignment in MEGA 5.2 (Tamura et al. 2011), both indicating that N12–N14 belonged to a congeneric species of *M. leucocephala*. However, we need more information to clarify this issue.

Meanwhile, we have acknowledged the fact that discrepancies between BLAST and BOLD identification results are due to slight differences in the sequence libraries and that the similarity between the results of the various methods, especially BLAST and the three additional methods, is due to the fact that they are mainly based on the same sequences.

Potential misidentifications and mislabelling in the inventory samples

Misidentification and mislabelling of samples are not rare in the previous studies (Pleijel et al. 2008; Ward et al. 2009; Yang et al. 2014) resulting in potential errors and confusion in later research. DNA barcoding provides a robust method to recognize potential cases of morphological misidentifications and mislabelling in inventory samples. For instance, it is highly likely that B16 and S51 were mislabelled as distantly related species due to the fact that all analytical methods generated consistent and unambiguous species classification with strong statistical support. Additionally, seven samples were highly likely to be misidentified from previous morphological analysis, involving species that are prone to be misidentification due to subtle morphological differences among sister species.

For example, B05 (*Falco rusticolus*) was empirically misidentified as *F. cherrug*. The morphological identification of raptors is difficult even for experts due to ambiguous identification criterion as well as morphological variation within species relative to age, gender, season and geographical distribution (Gamauf et al. 1998; Pearlstine and Thompson 2004). Morphologically *F. cherrug* cannot be clearly defined and shares overlapping phenotypic characteristics with *F. rusticolus* especially (Eastham et al. 2001). Other misidentifications (N7, N8, N31, N32) occurred between *Ciconia ciconia* and *C. boyciana* for similar reasons related to morphology. We also found misidentifications in several of our mammalian specimens, including *Trachypithecus hatinhensis* (S63, as *T. francoisi*) and *Nycticebus pygmaeus* (S49, as *N. coucang*). Unfortunately, most of the incongruent samples were incomplete making it impossible to re-identify them by the morphological method.

Factors impacting the application of DNA barcoding to conservation practices

To apply DNA barcoding to conservation and other practical or academic work as a routine method, we must acknowledge several factors. Sharing the same DNA barcode is the most notable issue, though it happens only occasionally. Employing multiple analytical methods

will have minimal impact on improving applicability in this case. Factors contributing to this issue include incomplete lineage sorting and introgression. For instance, we failed to identify B03 because of its sharing barcode with *A. virgatus* and *A. gularis*. To address such ambiguities, we advise including as many closely related species as possible or trying another DNA barcode marker.

Special attention should be paid to sibling species for they tend to be similar in morphology and genetic background, such as reported in Dixon (2012), thus, tend to be misidentified either by morphological approaches or DNA barcoding. The fewer consistencies detected between DNA barcoding and morphological identification in birds (78.72%) than that in mammals (87.18%) likely resulted from the higher number of sibling species included in our avian dataset.

The reference database also plays an important role in the applicability of DNA barcoding identification. Generally involving more barcodes of more species would improve the species assignment performance (Dincă et al. 2011). The identification failure of N12–N14 was mainly due to lacking DNA barcodes of *M. ibis* and other congeneric species. Meanwhile, the accurate information of downloaded barcodes is essential to obtain successful identification, including the species name and collection details. We found that the downloaded sequence NC016666 was labelled incorrect species information and similar cases had been mentioned in previous studies (Harris 2003; Mitchell 2008).

Conclusions

In this case study, DNA barcoding was applied to species identification in conservation practices based on endangered birds and large mammals in China. In order to better investigate the efficiency and feasibility of DNA barcoding identification in practical work, we used confiscated specimens and non-invasive samples and concealed species information during molecular identification. Forty-seven birds and 39 mammals were re-identified by DNA barcoding with sequential analyses of online primary species assignment, genetic distance, phylogenetic reconstruction, and diagnostic nucleotide methods. Eighty-two out of 86 queries were accurately assigned to a species level and four individuals to a genus level. Most of the DNA barcoding identifications (78.72% in birds and 87.18% in mammals) showed consistent species assignments with morphological classification. For those inconsistent with morphological classification, we identified seven potential morphological misidentifications and two cases of mislabelling with robust DNA barcoding evidence. The four samples identified to genus were mainly a result of sharing barcodes or lacking conspecific barcodes in the public database. Our survey showed a successful application of DNA barcoding to species authentication of unknown, incomplete samples in conservation practices and enriched the barcode database of endangered animals.

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Compliance with ethical standard

Conflict of interest The authors declare no conflict of interest.

References

- Ardura A, Linde AR, Moreira JC, Garcia-Vazquez E (2010) DNA barcoding for conservation and management of Amazonian commercial fish. *Biol Conserv* 143(6):1438–1443
- Armstrong K, Ball S (2005) DNA barcodes for biosecurity: invasive species identification. *Philos Trans R Soc Lond B* 360(1462):1813–1823
- Austerlitz F, David O, Schaeffer B, Bleakley K, Olteanu M, Leblois R, Veuille M, Laredo C (2009) DNA barcode analysis: a comparison of phylogenetic and statistical classification methods. *BMC Bioinform* 10(Suppl 14):S10
- Barber P, Boyce SL (2006) Estimating diversity of Indo-Pacific coral reef stomatopods through DNA barcoding of stomatopod larvae. *Proc R Soc Lond B* 273(1597):2053–2061
- Borisenko AV, Lim BK, Ivanova NV, Hanner RH, Hebert PD (2008) DNA barcoding in surveys of small mammal communities: a field study in Suriname. *Mol Ecol Resour* 8(3):471–479
- Côté IM, Green SJ, Morris JA Jr, Akins JL, Steinke D (2013) Diet richness of invasive Indo-Pacific lionfish revealed by DNA barcoding. *Mar Ecol Prog Ser* 472:249–256
- Dalton DL, Kotze A (2011) DNA barcoding as a tool for species identification in three forensic wildlife cases in South Africa. *Forensic Sci Int* 207(1–3):e51–e54
- DeSalle R, Egan MG, Siddall M (2005) The unholy trinity: taxonomy, species delimitation and DNA barcoding. *Philos Trans R Soc B* 360(1462):1905–1916
- Dincă V, Zakharov EV, Hebert PD, Vila R (2011) Complete DNA barcode reference library for a country's butterfly fauna reveals high performance for temperate Europe. *Proc R Soc Lond B* 278(1704):347–355
- Dixon A (2012) Conservation of the Saker Falcon *Falco cherrug* and the use of hybrids for falconry. *Aquila* 119:9–19
- Dove CJ, Rotzel NC, Heacker M, Weigt LA (2008) Using DNA barcodes to identify bird species involved in birdstrikes. *J Wildl Manag* 72(5):1231–1236
- Eastham CP, Nicholls MK, Fox NC (2001) Morphological variation of the Saker (*Falco cherrug*) and the implications for conservation. *Biodivers Conserv* 11(2):305–325
- Feng Y, Siu K, Wang N, Ng KM, Tsao SW, Nagamatsu T, Tong Y (2009) Bear bile: dilemma of traditional medicinal use and animal protection. *J Ethnobiol and Ethnomed* 5:1–9
- Gamauf A, Preleuthner M, Winkler H (1998) Philippine birds of prey: interrelations among habitat, morphology, and behaviour. *The Auk* 115(3):713–726
- González-Varo JP, Arroyo JM, Jordano P (2014) Who dispersed the seeds? The use of DNA barcoding in frugivory and seed dispersal studies. *Methods Ecol Evol* 5(8):806–814
- Gonçalves PF, Oliveira-Marques AR, Matsumoto TE, Miyaki CY (2015) DNA barcoding identifies illegal parrot trade. *J Hered* 106(S1):560–564
- Hajibabaei M, Janzen DH, Burns JM, Hallwachs W, Hebert PD (2006) DNA barcodes distinguish species of tropical Lepidoptera. *PNAS* 103(4):968–971
- Harris DJ (2003) Can you bank on GenBank? *Trends Ecol Evol* 18(7):317–319
- Hebert PDN, Cywinska A, Ball SL (2003) Biological identifications through DNA barcodes. *Proc R Soc Lond B* 270(1512):313–321
- Hebert PDN, Gregory TR (2005) The promise of DNA barcoding for taxonomy. *Syst Biol* 54(5):852–859
- Hebert PDN, Penton EH, Burns JM, Janzen DH, Hallwachs W (2004a) Ten species in one: DNA barcoding reveals cryptic species in the neotropical skipper butterfly *Astrartes fulgurator*. *PNAS* 101(41):14812–14817
- Hebert PDN, Stoeckle MY, Zemlak TS, Francis CM (2004b) Identification of birds through DNA barcodes. *PLoS Biol* 2:1657–1663
- Hubert N, Hanner R, Holm E, Mandrak NE, Taylor E, Burrige M, Watkinson D, Dumont P, Curry A, Bentzen P (2008) Identifying Canadian freshwater fishes through DNA barcodes. *PLoS ONE* 3(6):e2490
- Ivanova NV, deWaard JR, Hebert PDN (2006) An inexpensive, automation-friendly protocol for reconcealing high-quality DNA. *Mol Ecol Notes* 6:998–1002
- Jo H, Ventura M, Vidal N, Gim JS, Buchaca T, Barmuta LA, Jeppesen E, Joo GJ (2015) Disconcealing hidden biodiversity: the use of complementary monitoring of fish diet based on DNA barcoding in freshwater ecosystems. *Ecol Evol* 6(1):219–232
- Kerr KC, Stoeckle MY, Dove CJ, Weigt LA, Francis CM, Hebert PDN (2007) Comprehensive DNA barcode coverage of North American birds. *Mol Ecol Notes* 7(4):535–543
- Kimura M (1980) A simple method for estimating evolutionary rates of base substitutions through comparative studies of nucleotide sequences. *J Mol Evol* 16(2):111–120
- Kress WJ, García-Robledo C, Uriarte M, Erickson DL (2015) DNA barcodes for ecology, evolution, and conservation. *Trends Ecol Evol* 30(1):25–35

- Lau MW, Fellowes JR, Chan BPL (2010) Carnivores (Mammalia: Carnivora) in South China: a status review with notes on the commercial trade. *Mammal Rev* 40:247–292
- Li J, Zheng X, Cai Y, Zhang X, Yang M, Yue B (2015) DNA barcoding of Murinae (Rodentia: Muridae) and Arvicolinae (Rodentia: Cricetidae) distributed in China. *Mol Ecol Resour* 15(1):153–167
- Li J, Zhu Z (2014) Snow leopard poaching and trade in China 2000–2013. *Biol Conserv* 176:207–211
- MacKinnon J, Phillipps K, He F (2000) A field guide to the birds of China. Oxford University Press, Oxford
- Meier R, Shiyang K, Vaidya G, Ng PK (2006) DNA barcoding and taxonomy in Diptera: a tale of high intraspecific variability and low identification success. *Syst Biol* 55(5):715–728
- Mitchell A (2008) DNA barcoding demystified. *Aust J Entomol* 47(3):169–173
- Myers N, Mittermeier RA, Mittermeier CG, da Fonseca GAB, Kent J (2000) Biodiversity hotspots for conservation priorities. *Nature* 403(6772):853–858
- Nagy ZT, Sonet G, Glaw F, Vences M (2012) First large-scale DNA barcoding assessment of reptiles in the biodiversity hotspot of Madagascar, based on newly designed COI primers. *PLoS ONE* 7(3):e34506
- Oba Y, Ōhira H, Murase Y, Moriyama A, Kumazawa Y (2015) DNA barcoding of Japanese click beetles (Coleoptera, Elateridae). *PLoS ONE* 10(1):e0116612
- Pearlstine EV, Thompson D (2004) Geographic variation in morphology of four species of migratory raptors. *J Raptor Res* 38:334–342
- Pleijel F, Jondelius U, Norlinder E, Nygren A, Oxelman B, Schander C, Sundberg P, Tholleson M (2008) Phylogenies without roots? A plea for the use of vouchers in molecular phylogenetic studies. *Mol Phylogenet Evol* 48(1):369–371
- Quinto CA, Tinoco R, Hellberg RS (2016) DNA barcoding reveals mislabeling of game meat species on the US commercial market. *Food Control* 59:386–392
- Ratnasingham S, Hebert PDN (2013) A DNA-based registry for all animal species: the Barcode Index Number (BIN) system. *PLoS ONE* 8(7):e66213
- Saitou N, Nei M (1987) The neighbor-joining method: a new method for reconstructing phylogenetic trees. *Mol Biol Evol* 4(4):406–425
- Sambrook J, Russell DW (2001) Molecular cloning: a laboratory manual, 3rd edn. Cold Spring Harbor, New York
- Schipper J, Chanson JS, Chiozza F, Cox NA, Hoffmann M, Katariya V et al (2008) The status of the world's land and marine mammals: diversity, threat, and knowledge. *Science* 322:225–230
- Smith AT, Xie Y, Hoffmann RS, Lunde D, MacKinnon J, Wilson DE, Wozencraft WC, Gemma F (2010) A guide to the mammals of China. Princeton University Press, Princeton
- Tamura K, Peterson D, Peterson N, Stecher G, Nei M, Kumar S (2011) MEGA5: molecular evolutionary genetics analysis using maximum likelihood, evolutionary distance, and maximum parsimony methods. *Mol Biol Evol* 28(10):2731–2739
- Taylor HR, Harris WE (2012) An emergent science on the brink of irrelevance: a review of the past 8 years of DNA barcoding. *Mol Ecol Resour* 12(3):377–388
- van Velzen R, Weitschek E, Felici G, Bakker FT (2012) DNA barcoding of recently diverged species: relative performance of matching methods. *PLoS ONE* 7(1):e30490
- Vernooy R, Haribabu E, Muller MR, Vogel JH, Hebert PDN, Schindel DE, Shimura J, Singer GA (2010) Barcoding life to conserve biological diversity: beyond the taxonomic imperative. *PLoS Biol* 8(7):e1000417
- Ward RD, Hanner R, Hebert PDN (2009) The campaign to DNA barcode all fishes. FISH-BOL. *J Fish Biol* 74(2):329–356
- Ward RD, Zemlak TS, Innes BH, Last PR, Hebert PDN (2005) DNA barcoding Australia's fish species. *Philos Trans R Soc Lond B* 360:1847–1857
- Waugh J, Evans MW, Millar CD, Lambert DM (2011) Birdstrikes and barcoding: can DNA methods help make the airways safer? *Mol Ecol Resour* 11(1):38–45
- Weitschek E, Velzen R, Felici G, Bertolazzi P (2013) BLOG 2.0: a software system for character-based species classification with DNA barcode sequences What it does, how to use it. *Mol Ecol Resour* 13(6):1043–1046
- Wilson JJ, Rougerie R, Schonfeld J, Janzen DH, Hallwachs W, Hajibabaei M, Kitching IJ, Haxaire J, Hebert PDN (2011) When species matches are unavailable are DNA barcodes correctly assigned to higher taxa? An assessment using sphingid moths. *BMC Ecol* 11(1):18
- Wong EH-K, Hanner RH (2008) DNA barcoding detects market substitution in North American seafood. *Res Int* 41(8):828–837
- Yang C, Xiao Z, Zou Y, Zhang X, Yang B, Hao Y, Moermond T, Yue B (2014) DNA barcoding revises a misidentification on musk deer. *Mitochondr DNA* 0:1–8
- Yiming L, Wilcove DS (2005) Threats to vertebrate species in China and the United States. *Bioscience* 55(2):147–153

- Zhang L, Hua N, Sun S (2008) Wildlife trade, consumption and conservation awareness in southwest China. *Biodivers Conserv* 17(6):1493–1516
- Zhou Y, Meng X, Feng J, Yang Q, Feng Z, Xia L (2004) Review of the distribution, status and conservation of musk deer in China. *Folia Zool* 53(2):129–140